

BPE PROJECT
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Course Instructor: Ma'am Maheen Arshad

Submitted By:

Fatima Rathore (22i-2631)
Noor Fatima (22i-2610)
Musfirah Zunnon (22i-2692)
Saima Saleem (22i-2498)
Ayesha Shahid (22i-2483)

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Department of Software Engineering

FAST – National University of Computer & Emerging Sciences

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Executive Summary:

The creation of a BPMN project with an emphasis on XPDL file production and parsing is presented in this report. It entails building BPMN models for healthcare procedures while tackling difficulties with BPMN element identification and processing complexity. The Java-based parsing program has features like element counting and start/end event recognition, and it extracts BPMN elements from XPDL files. The paper closes with recommendations and insights on the advantages of business process engineering, as well as an open division of labor among group members.

Introduction:

In order to ensure alignment with business objectives and streamline operations, this project will investigate the use of XPDL (XML Process Definition Language) and BPMN (Business Process Model and Notation) in healthcare procedures. While XPDL facilitates the sharing of process models between various tools and systems, improving interoperability, BPMN offers a standardized graphical notation for modeling business processes, simplifying communication and documentation across stakeholders. Organizations may enhance processes, increase productivity, and guarantee better results by comprehending and utilizing BPMN and XPDL, especially in industries like healthcare where accuracy and efficacy are critical. This project focuses on advancing skills in process modeling and analysis using BPMN and XPDL, with tasks divided into Assignment 4 and the main project. Assignment 4 entails generating and parsing XPDL files, while the main project extends to dynamic process modifications, cycle time analysis, and manual creation of process variants from real-world data, aiming to deepen understanding and practical application in process engineering.

XPDL File Generation and Parsing:

Description of Bpmn

Private Process Model:

Patient activities are divided and represented using different bpmn symbols on the bases of time and date. Blood Test and X-ray scans are considered as sub processes and all activities are placed in the pool of Hospital.

Collaboration Process Model:

Hospital is divided into swim lanes according to Department and rooms related to activities given in event log file then each transition is shown between Laboratory, clinical room, Pharmacy, Operation Theatre, Radiology Room. The sequence flow followed is on the behalf of date and time.

Choreography Process Model:

There are two participants in every activity: doctor and patient. Patient Name is mentioned at the top and Doctor name is mentioned at the bottom. While the activity name is mentioned in center. The sequence flow followed is on the behalf of date and time.

Generation Process

- Firstly, the BPMN Models were created with proper notations using draw.io. Proper notations were used for gateways, tasks, events, and other elements.
- Then the BPMN Models were exported into XML. 'File' > 'Export' > 'XML'.

Development and Functionality of the Parsing Tool

The provided Java code represents a parser for XPDL (XML Process Definition Language) files, specifically tailored to extract various elements and characteristics of BPMN (Business Process Model and Notation) diagrams defined within XPDL files.

Technical Specifications:

Language and Libraries: The parser is written in Java and utilizes standard Java libraries for XML parsing and file handling.

Java library: `import org.w3c.dom.Document`

Parsing Methodology:

The parser first loads the XPDL file using DocumentBuilder and parses it into a DOM (Document Object Model) structure.

It then navigates through the DOM to extract specific elements and their attributes.

Element counts are calculated based on predefined criteria, such as styles and attributes associated with BPMN elements.

Functionality

Element Counting:

The parser counts various BPMN elements such as events, activities (tasks), gateways, artifacts, connecting objects (sequence flows, message flows, associations), and swimlanes.

It distinguishes between different types of tasks (user, service, script, manual, abstract, send) and gateways (exclusive, parallel, inclusive).

Start and End Events Detection:

It identifies start and end events based on specific styles associated with BPMN events.

Association, Sequence Flow, and Message Flow Counting:

Differentiates between these connecting objects based on their style attributes.

Handling Swimlanes:

Recognizes swimlanes and further differentiates between pools and lanes.

Error Handling:

The parser includes basic error handling for exceptions that may occur during XML parsing.

Challenges Faced and Solutions Implemented

Parsing Complexity:

XML parsing can be intricate, especially when dealing with complex BPMN diagrams. The solution uses the DOM parser, which provides a hierarchical representation of the XML document, making it easier to traverse and extract specific elements.

Identifying BPMN Elements:

BPMN elements are identified based on their styles and attributes, which can vary across different XPDL files. The solution employs criteria to accommodate variations in BPMN element definitions.

Results

```
<terminated> Main [Java Application] C:\Users\MEGA COMPUTER\.p2\pool\plugins\o
BPMN_Model_Elements:
Total_Events: 17
    Start_Events: 4
    Intermediate_Events: 6
    End_Events: 7
Total_Activities: 14
Tasks: 14
    User_Tasks: 1
    Service_Tasks: 4
    Script_Tasks: 0
    Manual_Tasks: 0
    Abstract_Tasks: 9
    Sub_Process: 0
    Send_Tasks: 0
Total_Gateways: 3
    Exclusive_Gateways_XOR: 0
    Parallel_Gateways_AND: 0
    Inclusive_Gateways_OR: 3
Total_Artifacts: 0
    Data_Objects: 0
    Groups: 0
    Annotations: 0
Total_Connecting_Objects: 30
    Sequence_Flows: 27
    Message_Flows: 3
    Associations: 0
Total_Swimlanes: 4
    Lanes: 4
    Pools: 0
```

Figure 1.1 File Name : Collaboration_Diagram_Milk_Collection_Phase

Adding of Calculation of Cycle Time Functionality

Assign Durations to Tasks:

Iterates through each mxCell element in the document.

For elements representing tasks (style containing "taskMarker"), a random duration is generated using the generateRandomDuration method from the TotalCycleTime class.

The duration is added as an attribute to the task cell (mxCell) along with an attribute to track if the task has been traversed (isTraversed).

Identify Split Gateways: Iterate through all elements in the BPMN diagram and identify elements that represent split gateways.

Calculate Total Outgoing Edges: For each split gateway, count the total number of outgoing edges.

Calculate Probabilities: Generate random probabilities for each outgoing edge such that the sum of probabilities equals 1.

Assign Probabilities: Assign probabilities to each outgoing edge of the split gateway.

Coding Description:

The function iterates through all elements in the BPMN diagram and identifies elements representing split gateways.

For each split gateway, it calculates the total number of outgoing edges by traversing through all edges in the document.

Random probabilities are generated for each outgoing edge using the calculateProbabilities function.

The assignProbabilitiesToOutgoingEdges function assigns these probabilities to each outgoing edge of the split gateway.

Methodology for Calculating Cycle Times:

The calculateCT function calculates the cycle time of a sequence of tasks in a BPMN diagram.

It recursively traverses through the tasks and gateways, accumulating the durations of tasks along the path.

For gateways, it considers different scenarios:

Split Gateways: It calculates the maximum duration among the branches.

Join Gateways: It calculates the duration considering all incoming paths.

Exclusive Gateways: It considers the duration of the path that is chosen based on the conditions.

Probabilities assigned to sequence flows from split gateways are used to calculate the expected duration of each path.

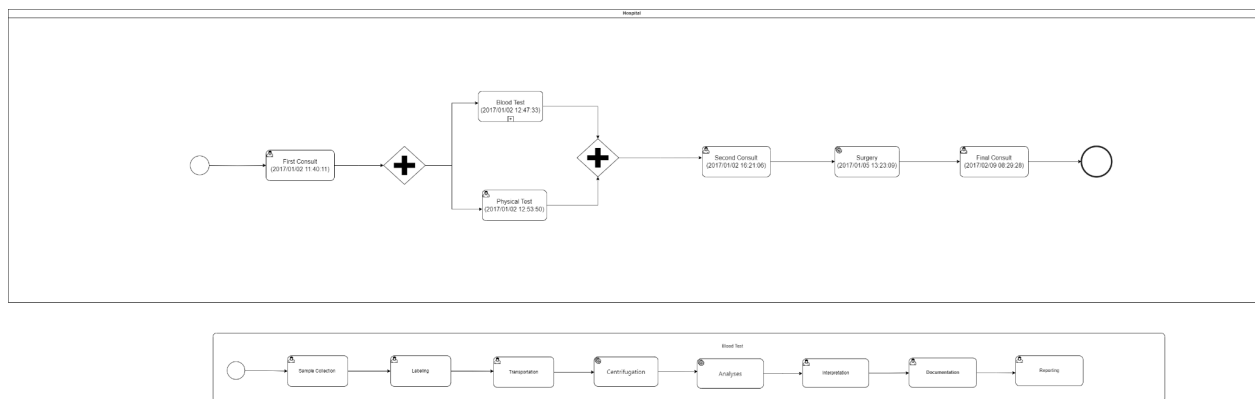
Creation Of Process Variants

The creation of BPMN Models involves different steps.

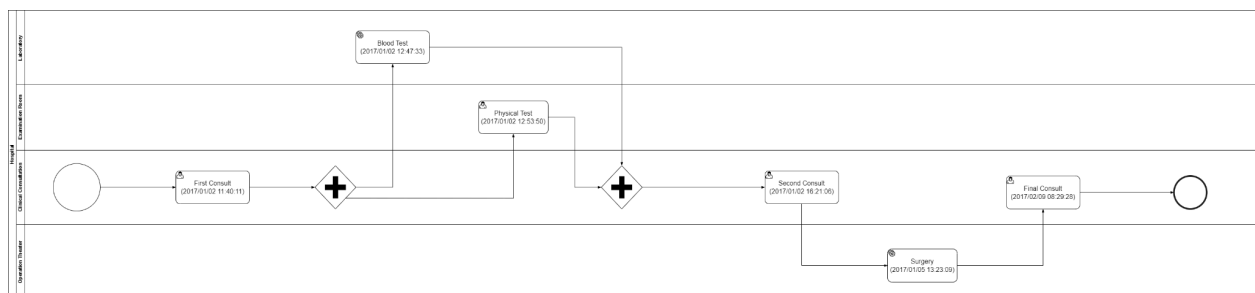
- Use Java code to sort the files of the event log based on a patient number.
- After sorting the file, different patients where the sequence of activities and tasks was different were identified.
- Then, the different types of tasks or activities that are performed were identified. This may include tasks such as X-ray scans, blood tests, physical tests and surgery etc.
- BPMN model types such as private process model, collaboration model and choreography model were kept under consideration and by using all the proper notation, BPMN models are made.

Process Variants:

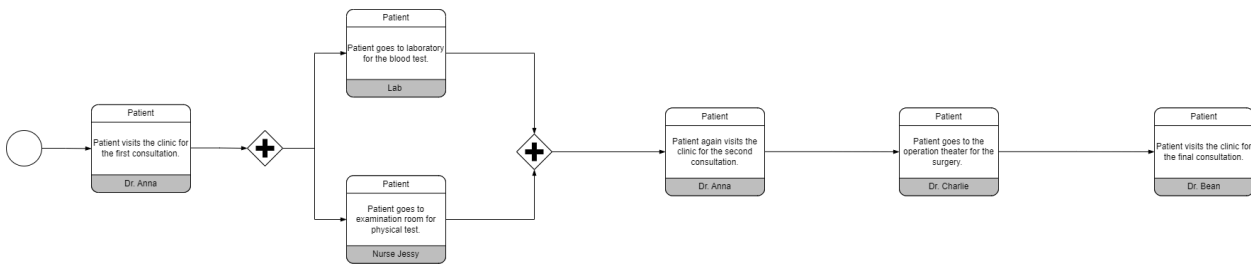
Patient 0: Private Process Model



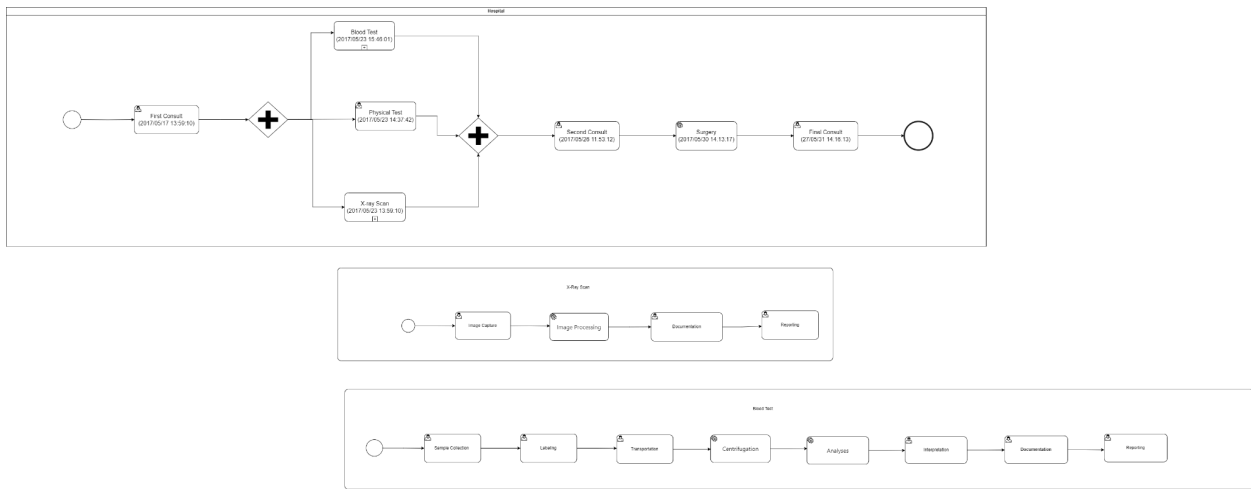
Patient 0: Collaboration Model:



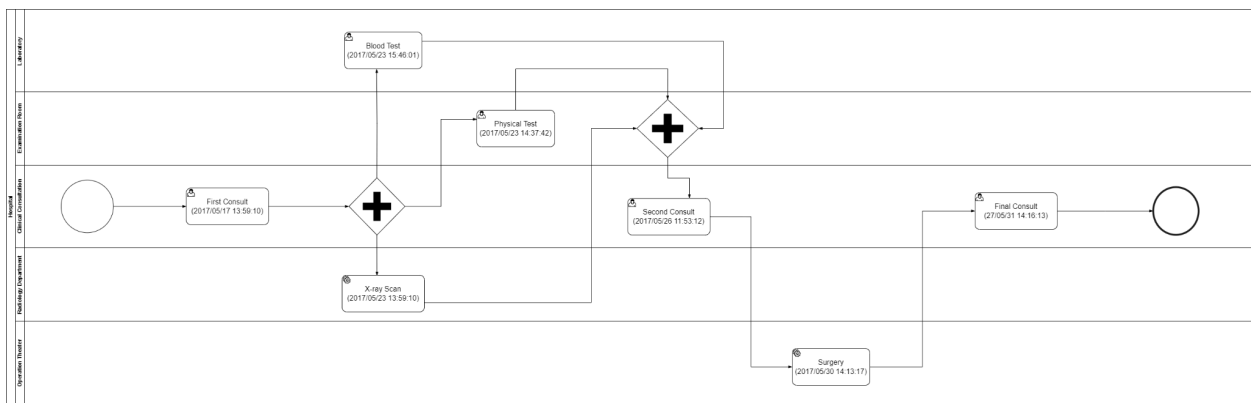
Patient 0: Choreography Model:



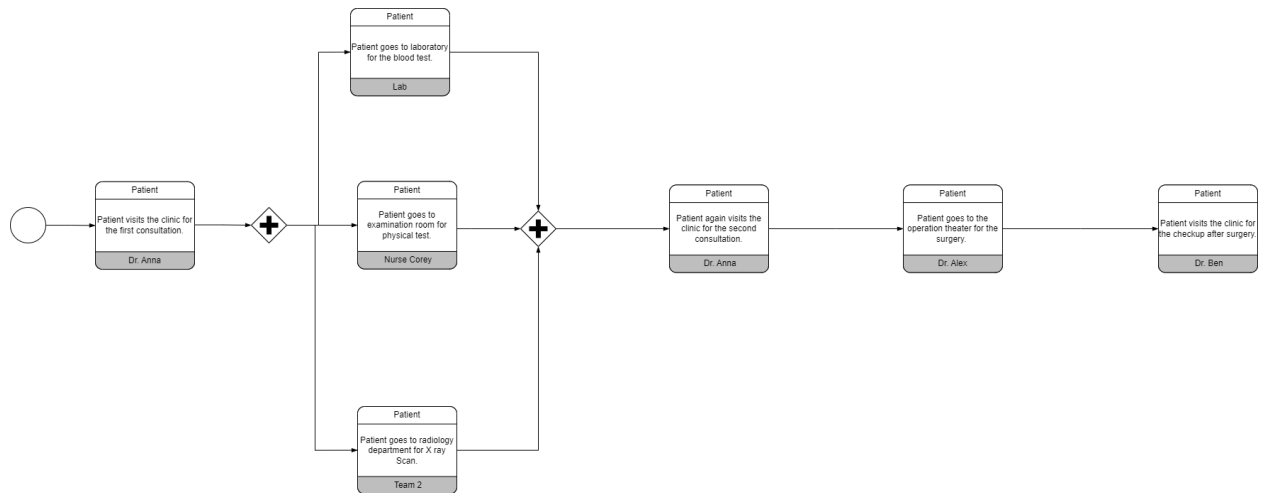
Patient 98: Private Process Model



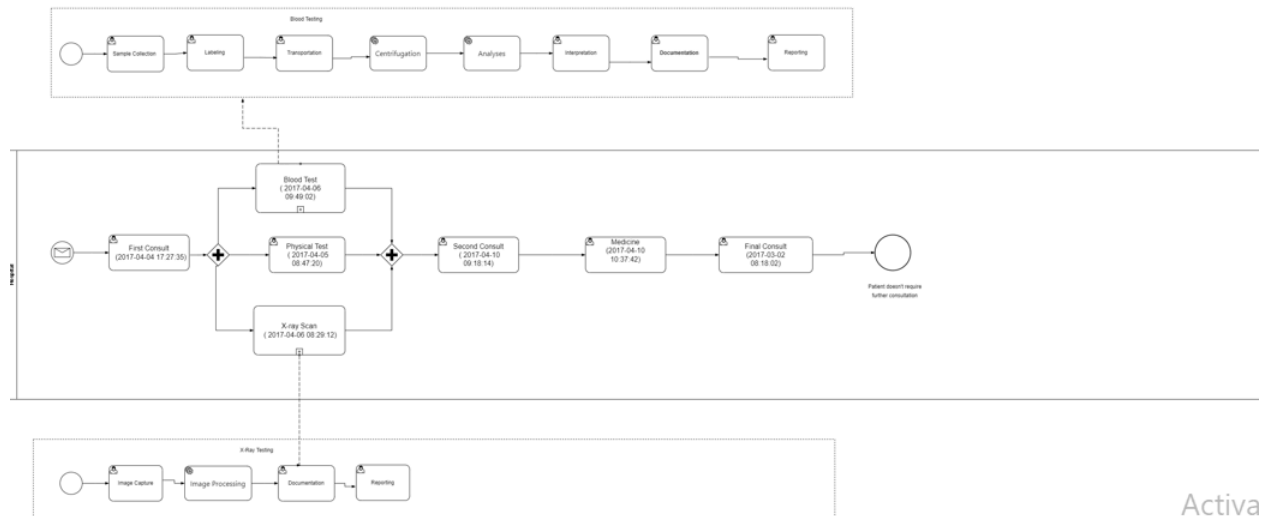
Patient 98: Collaboration Model:



Patient 98: Choreography Model:

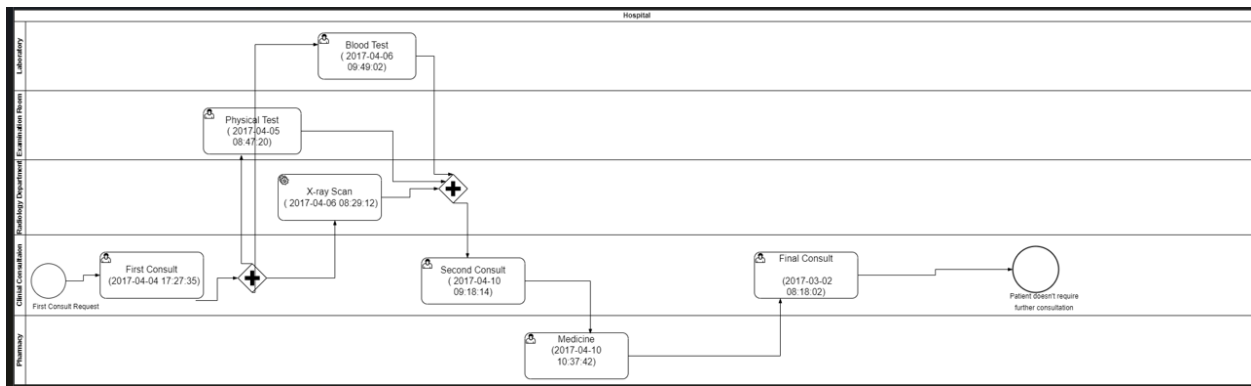


Patient 65: Private Model:

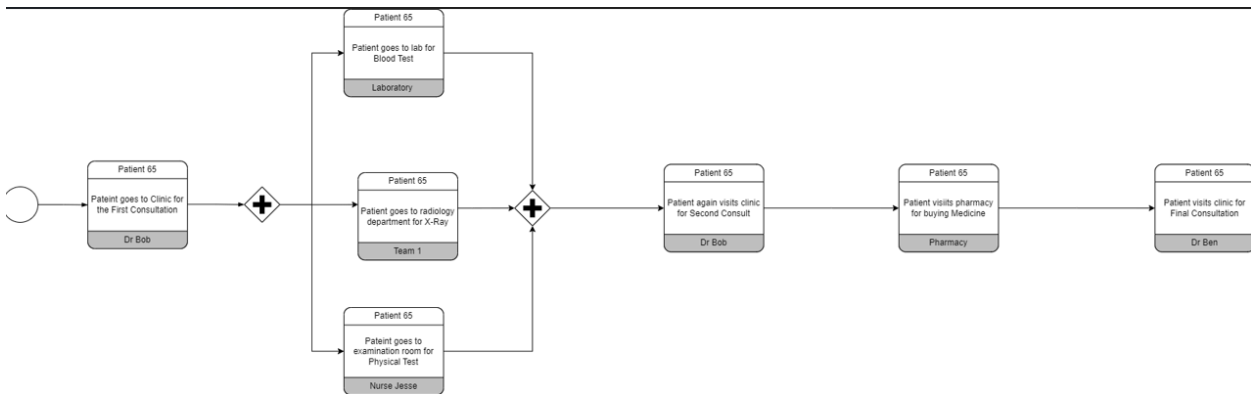


Activa

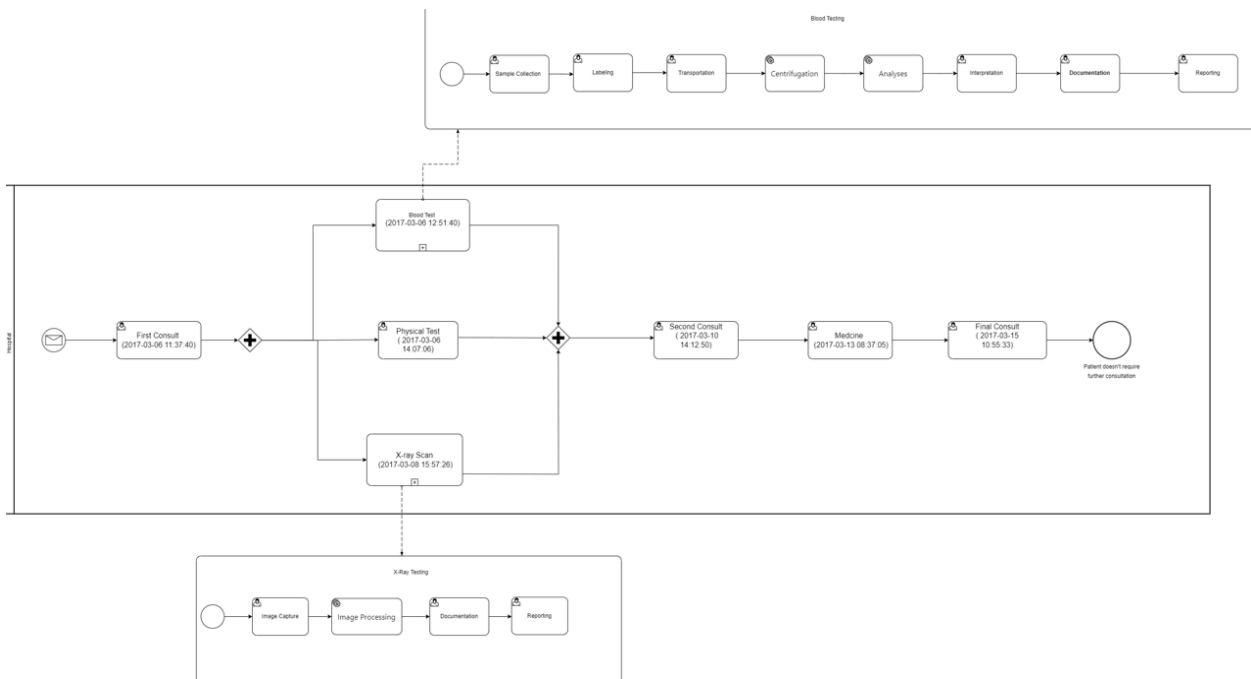
Patient 65: Collorabation Model:



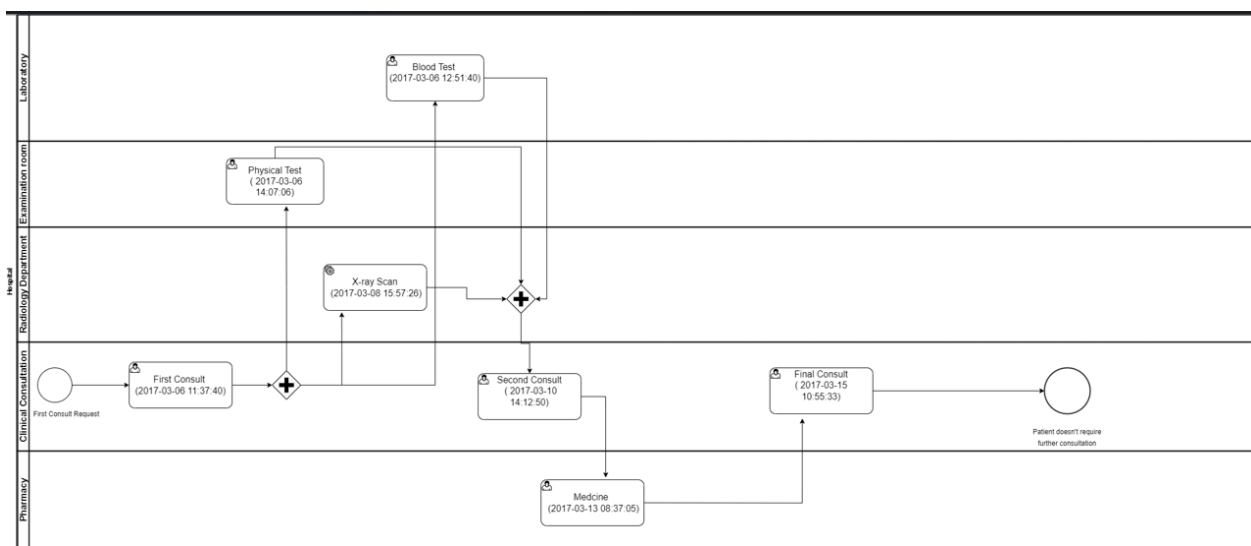
Patient 65: Choreography Model:



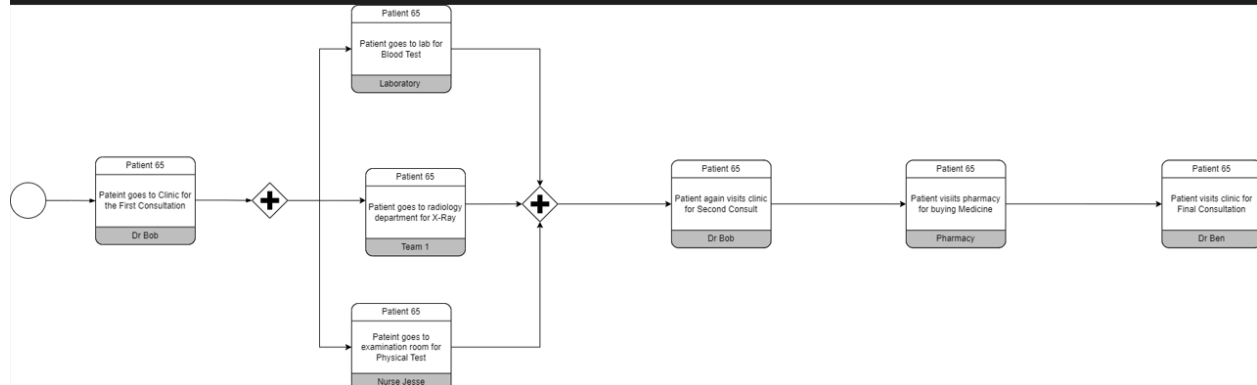
Patient 44: Private Model:



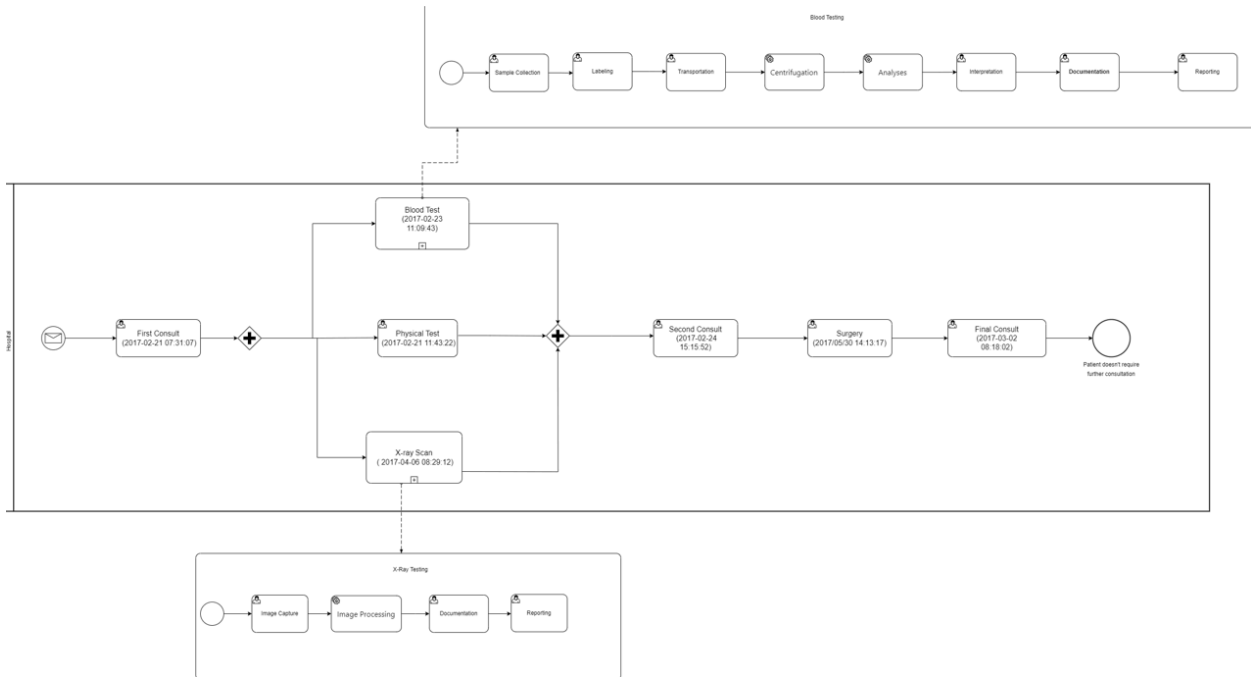
Patient 44: Collaboration Model:



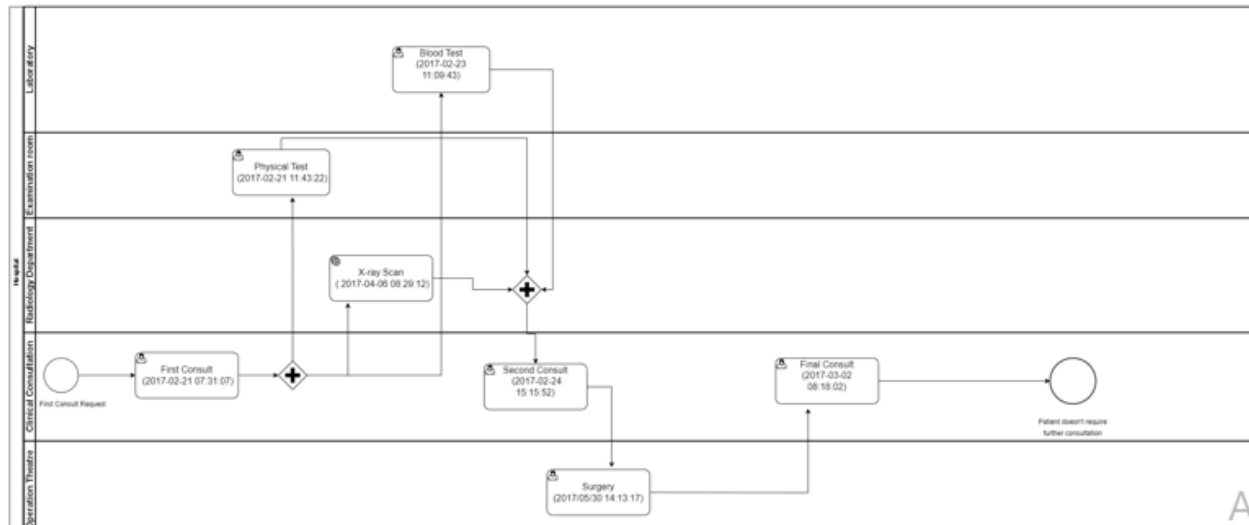
Patient 44 Cherography:



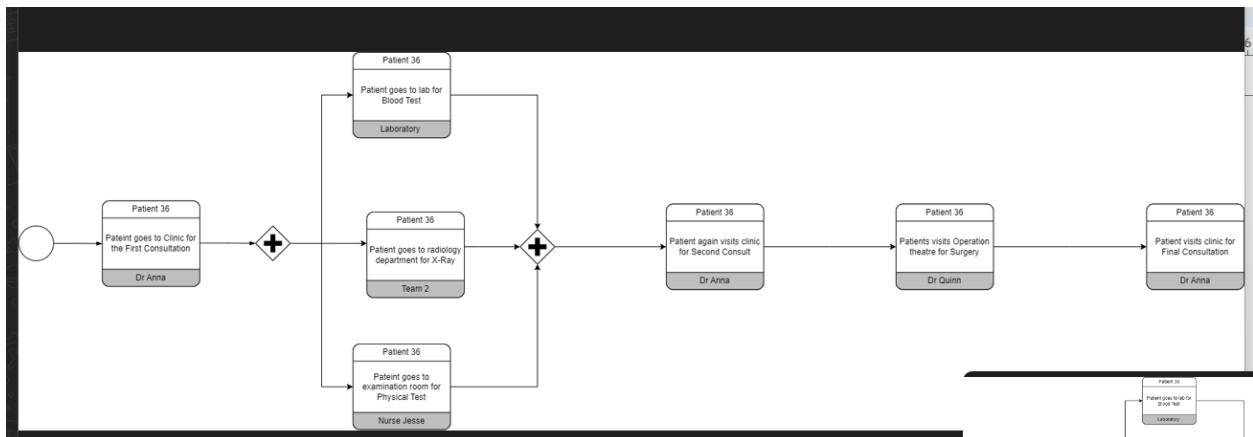
Patient 36: Private



Patient 36: Collaborative



Patient 36 Choreography



Conclusion:

Learning and Outcomes:

However, despite these challenges, some key learnings and outcomes can be from business process engineering projects. The organizations can benefit from improved efficiency, cost savings, better customer satisfaction, and increased competitiveness by optimizing their business processes. Additionally, successful process engineering efforts can foster a culture of continuous improvement and innovation within the organization.

Improvements:

To ensure successful business process engineering:

1. Engage stakeholders at all levels, implement robust change management, document processes thoroughly, align technology with business goals, establish data governance, foster cross-functional collaboration, and institute measurement mechanisms for continuous improvement.
2. Foster a culture of collaboration, innovation, and learning to drive sustainable process enhancements and adaptability to evolving business landscapes.

Work Division:

Group Member	Work Division
Noor Fatima	XPDL Parser Cycle Time Calculation
Fatima Rathore	
Musfirah Zunnoon	
Saima Saleem	Creation of Variants
Ayesha Shahid	